

- Q.28 Difference between Newtonian Fluid and Non Newtonian Fluid?
- Q.29 Explain about the working of Rotameter with neat clean diagram.
- Q.30 Explain about dimensionless numbers and its significance.
- Q.31 Derive the equation for Hagen Poiseuille's equation?
- Q.32 What is the purpose of colour coding used in industry?
- Q.33 Discuss about Properties of flow?
- Q.34 How does cavitation affect the performance of a pump?
- Q.35 Explain about the transportation of various fluids.

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 With construction and working, explain complete about of Reciprocating pump
- Q.37 Explain the working principle of a venturimeter in detail. Include the key components, flow dynamics, and the equations used to calculate flow rate.
- Q.38 Define manometer, write about various types of manometers. Explain how a simple U-tube manometer works.

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**3rd Sem / Chem, P&P, Chem Engg. (Spl. Paint Tech.)  
Chem Engg. (Spl. Polymer Engg.)**

**Subject:- Fluid Flow**

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which type of flow is characterized by smooth and orderly movement of fluid particles in parallel layers?
- a) Turbulent flow      b) Laminar flow  
c) Compressible flow      d) Irrotational flow
- Q.2 Which of the following devices is commonly used for measuring the flow rate of liquids in pipelines?
- a) Venturi meter      b) Pitot tube  
c) Orifice plate      d) Bourdon gauge
- Q.3 Bernoulli's principle describes the relationship between:
- a) Pressure and velocity in a fluid  
b) Temperature and density in a fluid  
c) Viscosity and turbulence in a fluid  
d) Mass and volume of a fluid
- Q.4 A centrifugal pump works based on the principle of:
- a) Axial flow  
b) Positive displacement  
c) Centrifugal force  
d) Bernoulli's principle

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- Q.5 Which type of flow has constant velocity and properties at any given point in the flow field?
- Laminar flow
  - Compressible flow
  - Turbulent flow
  - Steady flow
- Q.6 To prevent cavitation in a pump, which parameter should be carefully considered and controlled?
- Fluid temperature
  - Fluid viscosity
  - Fluid density
  - NPSH (Net Positive Suction Head)
- Q.7 Which type of valve is commonly used to regulate the flow of fluids by partially obstructing the flow path?
- Check valve
  - Gate valve
  - Ball valve
  - Globe valve
- Q.8 Which flow measurement devices utilizes the deflection of a moving fluid jet to measure flow rate?
- Venturi meter
  - Magnetic flowmeter
  - Orifice meter
  - Pitot tube
- Q.9 The reynold number is  $Re < 4000$  for
- Laminar flow
  - Transitional flow
  - Turbulent flow
  - All of above
- Q.10 In compressible flow, which property of the fluid can change significantly with changes in pressure and velocity?
- Viscosity
  - Density
  - Temperature
  - Surface tension

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## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the SI unit for flow rate?
- Q.12 Define the terms “compressibility” in the context of fluids.
- Q.13 Why do we measure flow?
- Q.14 Write full form of NPSH.
- Q.15 Can you name a non-Newtonian fluid?
- Q.16 Give an example of incompressible flow.
- Q.17 Write formula of hydrostatic law
- Q.18 Write one application of compressor?
- Q.19 Name one advantage of an orifice meter.
- Q.20 What is uniform Flow?

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between positive displacement and centrifugal pumps.
- Q.22 Briefly explain Bernoulli’s principle and its implications for fluid flow.
- Q.23 What is valves and explain any two types of valves.
- Q.24 Differentiate between blower and compressor.
- Q.25 Difference between steady and unsteady, rotational and irrotational flow.
- Q.26 Describe about pascal law with its derivation.
- Q.27 Explain about various types-Friction Losses in Pipes.

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